FitBuddy – FastAPI Backend Implementation

Team / Overview / Workspace / Kanban

Story Duration: 1h 30m

1. Implement the FastAPI Backend to Manage Routing and User Input Processing

The FastAPI backend handles application routing, processes user inputs, connects with Gemini-powered functions, and returns AI-generated responses to the frontend.

2. Define Routes in FastAPI

All routing logic is handled in **routes.py**. Each route is linked to the corresponding function developed in Activity 2.1.

3. Process User Input

HTML Form: Created in **index.html**.

Users enter the following information:

- `username`
- `user_id`
- `age`
- `weight`
- `goal`
- `intensity`

FastAPI captures the inputs using **Form(...)** parameters.

Pydantic schemas **UserInput** and **FeedbackRequest** are used to validate the structure of the submitted data.

4. Integrate Gemini API Calls

Each route handler connects directly to the respective Gemini-powered function.

```
/generate-workout
    → generate_workout_gemini() for workout plan

generate_nutrition_tip_with_flash()
    → Generates a nutrition tip

/submit-feedback
    → Retrieves the existing plan
    → Sends the original plan + feedback to update_workout_plan()
```

5. Frontend Integration

All AI-generated responses are returned cleanly and injected into the frontend templates for display.